

What Helps When Telemetry Lies?

Four Pre-Registered Tests of Evidence Acquisition under Source Compromise

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Abstract—A threat hunter with a fixed budget must choose which telemetry sources to read, knowing that some of them may be controlled by the attacker. Acquisition rules for this setting are rarely compared at equal spend, against attackers they were not designed for, with the comparison fixed before the data exist. We run four pre-registered experiments, each changing one part of a Bayesian acquisition policy while the shared posterior, the terminal decision, the horizon and the spend stay fixed. Three run in a simulator over procedurally generated worlds (201,600, 499,200 and 537,600 policy-episodes) and one on a public industrial-control dataset. Two ideas fail their gates. Valuing integrity probes by their decision consequences adds nothing over hypothesis information under the same model (relative regret reduction 0.02, 95% CI $[-0.02, 0.06]$). Randomised and minimax acquisition do not beat a non-robust policy on a held-out attacker (-0.19 and -0.02). One passes every gate: in simulation, scoring evidence by hypothesis information under a joint posterior over the hypothesis and every source’s compromise state reduces regret on a held-out attacker by 0.27 $[0.23, 0.30]$ relative to reliability-weighted information gain and by 0.22 $[0.19, 0.26]$ relative to joint-state information gain. The held-out attacker lies partly in the direction the defender’s model assumes. The gain is large against cover-up and decoy attackers, small against misattribution, and against an attacker that adapts to the defender’s queries it survives only relative to reliability weighting; there random acquisition is best. A replication on the HAI dataset was invalid by its own pre-registered check, because the mapped evidence did not beat a decision from the prior alone, so the positive result is simulator-only.

Index Terms—threat hunting, adversarial telemetry, active hypothesis testing, value of information, pre-registration, negative results

I. INTRODUCTION

Attackers tamper with the evidence defenders rely on. Log deletion, forwarder sabotage and sensor spoofing are catalogued techniques (MITRE ATT&CK T1562, “Impair Defenses” [1]), and holding a process variable at a stale value is a standard attack on industrial control systems [2], [3]. A hunter who investigates an alert under a budget therefore faces two coupled questions: which source to read next, and how far to believe it.

Each piece of this problem has a long history. Choosing the next measurement by expected entropy reduction over a joint fault model goes back to the General Diagnostic Engine [4]; decision-theoretic troubleshooting values tests by their effect on the repair decision [5]; active sequential hypothesis testing gives asymptotically optimal sensing rules [6]; and Byzantine detection studies inference when some sensors lie [7]–[9].

Recent work combines a belief over compromised sensors with active probing in cyber-physical control [10] and learns cost-aware sensor scheduling with a Byzantine detector [11].

What is missing is a controlled answer to a practical question: *which part of such a policy actually reduces decision errors when sources lie?* A new policy usually differs from its baseline in several ways at once (belief model, objective, look-ahead, cost handling), is evaluated against the attacker its model assumes, and is compared at unequal spend. Any of these can manufacture an apparent gain.

We address this with an ablation discipline rather than a new method. Every policy in this paper shares the same exact posterior, the same Bayes terminal decision and exactly the same spend; policies differ only in how they score the next action. Each experiment changes one part, names its comparators, fixes a practical threshold (a 10% relative cut in regret) and its gates in a committed protocol, and is tested on an attacker written after the policies were frozen.

Contributions.

- 1) A spend-matched, pre-registered testbed for evidence acquisition under source compromise: procedurally generated worlds, five attacker families of which three were each held out from one experiment’s design, and frozen gate code (Sec. IV).
- 2) Two pre-registered negative results. Valuing source-integrity probes by their decision consequences adds no robust gain over hypothesis information under the same joint model, and neither randomised nor minimax acquisition beats the non-robust comparator on a held-out attacker (Sec. V-A).
- 3) One scoped positive result. At equal objective, horizon and cost handling, replacing a reliability-weighted look-ahead model with the joint model over hypothesis and source states reduces regret by 0.27 on a held-out attacker, and replacing the joint-entropy objective with hypothesis entropy reduces it by 0.22. We report where both gains shrink or vanish (Sec. V-B).
- 4) A real-data attempt on the HAI industrial-control dataset that failed its own validity check, with the reasons, so that the positive result is reported as simulator-only (Sec. VI).

Scope. All positive evidence comes from one simulator family and one world distribution, with policies and attackers written by the same author. We do not claim the joint belief

model as new; it is classical [4], [5], [8]. The contribution is controlled evidence about which part matters, and where it stops mattering.

II. RELATED WORK

Diagnosis and troubleshooting. GDE selects measurements by minimum expected entropy over component states [4]; Bayesian-network test selection does the same at scale [12]; decision-theoretic troubleshooting values tests by their effect on the repair decision [5]. Faults there are not strategic, and observations of healthy components are trusted. Our joint-state objective J is the GDE criterion and our decision-value probe P follows the troubleshooting criterion.

Sensing with adversarial sources. Byzantine detection studies fusion when a fraction of sensors lie, through error exponents and game equilibria with all sensors observed [7], [9]. Soltanmohammadi et al. infer node identity jointly with the hypothesis by EM [8], the closest inference-side match to our belief model, but without a choice of which source to read. Andersson and Dán maintain a Bayesian network over sensor attack states and choose a control input that maximises a KL divergence between attack hypotheses, with the goal of control recovery [10]. Zhong et al. learn sensor scheduling with probing costs and a separate Byzantine detector [11]. MATE estimates per-agent trust as a hidden state for fusion [13]. In industrial control, residual-based detectors and their limits against stealthy attacks are well studied [3]; our HAI probe is a simple instance. None of these ablates the belief model against the objective at equal spend, or evaluates on attackers written after the policy.

Active learning with unreliable oracles. Proactive learning selects among imperfect, cost-varying oracles [14]; cost-aware active learning divides information by cost [15], [16]. Oracle noise there is not strategic.

Security operations. POMDP and optimal-stopping formulations plan sensing, prevention and response under noisy alerts [17]–[19], but treat alert noise as non-strategic. Forward-secure and tamper-evident logging [20], [21] protect the integrity a probe would check; provenance triage [22] and hunting under anti-forensics [23] motivate our compromised-source model. These works do not choose evidence decision-theoretically under source compromise.

III. PROBLEM AND POLICIES

Operational picture. A hunter triages an alert on a host. The sources are the telemetry a SOC would pull: EDR, Sysmon, authentication logs, DNS, proxy, NetFlow, cloud audit logs, mail gateway. A query pulls evidence from one source; it can be forged if the attacker controls that source’s sensor or forwarder. A probe is an integrity check on one source: inject a canary event and verify it arrives intact, check the forwarder’s heartbeat or hash chain, or cross-check the source against an independent sensor. The final decision is to dismiss the alert, contain the host or block egress.

State. A hidden hypothesis $H \in \{\text{benign}, \text{intrusion}_1, \text{intrusion}_2\}$ and, for each of K sources,

a state $S_q \in \{\text{healthy}, \text{degraded}, \text{compromised}\}$. A healthy source reports the true hypothesis with probability acc_q ; a degraded one reports uniformly; a compromised one, in the defender’s model, forges its report with probability 0.9: a cover-up (“benign”) under an intrusion, a false flag under benign. A probe returns PASS or FAIL with state-dependent rates. Compromise is less likely under benign activity: $P(S_q=C \mid \text{benign}) = \kappa P(S_q=C \mid \text{intrusion})$. Sources are conditionally independent given (H, S) , which ignores shared failure modes such as one host agent feeding several sources (Sec. VII).

Decision and regret. After the budget is spent, the defender takes $D^* = \arg \max_D \mathbb{E}[U(D, H) \mid E]$: 0 if correct, $-L_{\text{miss}}$ for dismissing an intrusion, $-0.5L_{\text{miss}}$ for the wrong containment, -1 for a false alarm. Regret is $U(D^*(H), H) - U(D, H)$ divided by L_{miss} , so it lies in $[0, 1]$.

Shared machinery. Every policy sees the same exact posterior $P(H, S_1, \dots, S_K \mid E)$, and the simulator, not the policy, applies the same Bayes decision. Costs are integers with a fixed budget, and every policy must act while it can afford an action, so every policy spends exactly the budget in every episode (checked in every run). Policies never see the true hypothesis, source states or attacker identity; leakage tests enforce this.

Policies differ only in scoring. A score has three parts: the *model* used for look-ahead, the *objective* credited per outcome, and the *horizon*. With horizon 2 and cost-awareness, $\text{score}(a) = \max(V_1(a)/c_a, \max_b V_2(a, b)/(c_a + c_b))$. The four policies that matter are:

- R (P3_RHAT_COST_LA2): reliability-weighted look-ahead. A query of q is scored as $\hat{r}_q p_{\text{nom}} + (1 - \hat{r}_q) \text{uniform}$, with $\hat{r}_q$ the current posterior probability that q is healthy. Objective: entropy reduction of H .
- C (P_JOINT_HEIG_COST): exact joint look-ahead; objective: entropy reduction of H .
- J (P_JOINT_EIG_COST): exact joint look-ahead; objective: entropy reduction of (H, S) , the GDE criterion.
- P (P_PROBE): exact joint look-ahead; objective: expected gain in the Bayes-decision utility (value of information), falling back to H -information on VoI plateaus.

All four use horizon 2 and divide by cost. C and J differ only in the objective. C and R differ only in the look-ahead model, but that model difference is a package of four choices: R treats a compromised source as symmetric noise rather than as a cover-up forger; it drops the coupling κ between H and S ; in its look-ahead a query updates H but not the queried source’s state, so repeated reads of one source count as independent; and it credits a probe with no immediate H -information. Our experiments do not separate these four. References include random acquisition, nominal EIG that ignores cost and compromise (P3_NOM), and, in E2, two robust variants of J .

IV. EXPERIMENTAL SETUP

Worlds. Each unit of analysis is a world drawn from a frozen distribution: $K \sim U\{5..8\}$ sources;

$P(\text{benign}) \sim U(0.30, 0.80)$, the rest split evenly; source accuracy $U(0.45, 0.95)$; query cost $U\{1..4\}$ with one source forced to 1; probe cost $\max(1, \text{round}(rc_q))$ with $r \sim \log-U(0.25, 2)$; compromise prior $\pi \sim U(0.02, 0.40)$ with per-source jitter $U(0.5, 1.5)$, capped at 0.60; benign coupling $\kappa \sim U(0, 1)$; degradation prior $U(0, 0.15)$; probe detection of compromise $U(0.5, 0.99)$, of degradation $U(0.6, 0.99)$, false-fail $U(0, 0.1)$; loss ratio $L_{\text{miss}} \sim \log-U(1, 20)$; budget 12. No world is hand-built. The ranges are deliberately wide to map behaviour across regimes; they are not calibrated to any SOC.

Attackers. An attacker controls $B \in \{1, 2, 3\}$ sources and is isolated from the policies. Under benign activity every family is present with probability κ , matching the defender’s coupling assumption. Five families: *cover-up* (the design family, matching the defender’s model); *misattribution* (targets the most trusted sources and pushes the wrong containment; held out in E1); *trust-harvest* (adaptive: honest until queried, then takes the source over and lies low after a probe); *decoy-migrate* (a sticky decoy class, migrates when probed; held out in E2); *collude-cheap* (controls the cheapest sources, which all tell the same lie, a cover-up or a wrong intrusion class with equal probability under an intrusion; held out in E3). A clean condition has no attacker. Within a world every policy faces the same truths and random numbers (common random numbers), 12 episodes per (world, condition).

Endpoint and statistics. For candidate X and comparator Y , $\bar{y}_w$ is a world’s mean normalised regret over the primary cells and $\text{RRR} = 1 - \sum_w \bar{y}_w(X) / \sum_w \bar{y}_w(Y)$. The main contrast gates pass only if $\text{RRR} \geq 0.10$ and the lower end of a two-sided percentile paired bootstrap interval over worlds (10,000 resamples, fixed seed) exceeds 0: 95% in E1, E3 and E4 (one-sided $\alpha = 0.025$), 97.5% in E2 (two candidates, Bonferroni). The 0.10 bar applies to the point estimate; the interval tests superiority over 0, not over 0.10. Other gates are not RRR gates: E1’s comparison with C used a Holm-adjusted bootstrap $p < 0.05$; E1 also gated on probe targeting and an ablation; every experiment gated on clean-world non-regression, exact spend and leakage tests. The number of worlds came from a blinded calibration on separate seeds that output only the comparator’s mean and the spread of the difference. E1 needed 139 worlds and ran 140; E2 and E3 needed 531 and 432 and ran the pre-registered cap of 400, so they are slightly under-powered for an RRR of exactly 0.10.

Validity checks. Each experiment has one. In E1–E3 the attacker must raise the regret of nominal EIG over clean worlds (V-sim). In E4 nominal EIG must beat a decision taken from the prior alone (V-real). If the check fails, the run is *invalid* and no gate decides anything, whatever its value.

Pre-registration. For each experiment the protocol, gates, configuration, gate code and tests were committed before any calibration or main run; the runners refuse a dirty tree, a changed configuration or an existing output, and the analysis scripts refuse to overwrite a verdict. Table I lists the experiments and their verdicts.

TABLE I
THE FOUR PRE-REGISTERED EXPERIMENTS AND THEIR VERDICTS.

Part tested	Comparators	Units	Verdict
E1 decision-value probing (P)	J ; R , C	140 worlds	KILL
E2 randomised / minimax J	J ; random	400 worlds	KILL
E3 H -info under joint model (C)	R and J	400 worlds	PROCEED
E4 E3 on real data (HAI)	R and J	100 episodes	INVALID

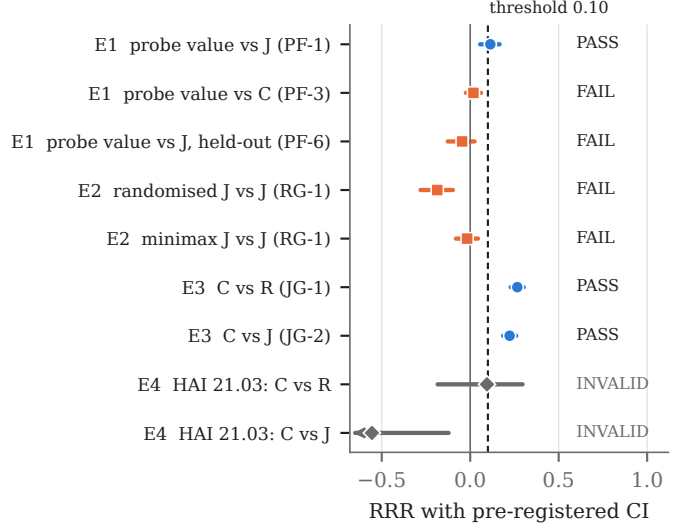


Fig. 1. Primary contrasts of all four experiments: RRR (marker) and pre-registered interval (bar) against the 0.10 threshold. “PASS”/“FAIL” is the gate on that row; E1 passed its headline gate but failed three others and was killed (Table I). E4 rows are shown for completeness; the run was invalid. Bars below -0.65 are truncated (arrow).

V. RESULTS

Fig. 1 and Table II show every primary contrast with its pre-registered interval. V-sim passed in E1–E3 and V-real failed in E4; spend matching and leakage gates passed in all four.

A. Negative results

E1: decision-valued probing. P beat J on the pooled attacker families (RRR 0.11 [0.06, 0.17]), but the experiment was killed by three gates. Against C , which uses the same joint model with plain H -information, the gain vanished (0.02 [−0.02, 0.06]). On the held-out misattribution attacker it was negative (−0.05 [−0.13, 0.02]). The pooled gain came from the design attacker, whose behaviour matches the defender’s compromised-source model (0.49 [0.41, 0.56]). Probing was also untargeted: only about a third of P ’s probes went to sources the posterior considered suspicious, against a pre-registered floor of one half.

E2: robust acquisition. E1’s exploratory data suggested that model-based policies fail when the attacker does not match their model. E2 tested two fixes, each changing J in one way: sampling actions in proportion to their score, and scoring by the worst case over three compromised-source models. On a new held-out attacker neither beat J : randomisation lost clearly (−0.19 [−0.28, −0.10]) and the minimax policy tied

TABLE II
PRIMARY CONTRASTS. RRR [INTERVAL]: 95% FOR E1, E3, E4; 97.5% FOR E2. “POOLED FAMILIES” ARE THE THREE E1 ATTACKER FAMILIES (COVER-UP, MISATTRIBUTION, TRUST-HARVEST).

Contrast	Gate	RRR [interval]	Result
E1 P vs J , pooled families	PF-1	0.114 [0.056, 0.166]	pass
P vs C , pooled families	PF-3	0.018 [−0.025, 0.061]	fail
P vs J , held-out misattrib.	PF-6	−0.046 [−0.127, 0.025]	fail
E2 Rand. J vs J , held-out	RG-1	−0.186 [−0.281, −0.097]	fail
Minimax J vs J , held-out	RG-1	−0.017 [−0.081, 0.044]	fail
E3 C vs R , held-out	JG-1	0.266 [0.229, 0.303]	pass
C vs J , held-out	JG-2	0.223 [0.186, 0.260]	pass
E4 C vs R , HAI 21.03	HG-1	0.095 [−0.184, 0.296]	n/a
C vs J , HAI 21.03	HG-2	−0.556 [−1.438, −0.122]	n/a

(−0.02 [−0.08, 0.04]) while losing on the known families. Both wrappers were built on J , the comparator E1 had named; robust versions of C were not tested.

B. Scoped positive result

Origin, disclosed. In exploratory tables after the E1 and E2 verdicts, C had lower regret than the one-step reliability-weighted policy (by 28% over E1’s attacker cells and 55% on E2’s held-out attacker) and than J (by 52% on E2’s held-out attacker), with near-zero gain against misattribution and trust-harvest. E3 tested this prospectively against the stricter two-step R , on new world seeds and a new held-out attacker (collude-cheap) written before any E3 run. No policy, threshold or world parameter was changed.

E3: joint belief. Every gate passed. On collude-cheap, C reduced regret by 0.27 [0.23, 0.30] relative to R (Cliff’s $\delta = 0.23$) and by 0.22 [0.19, 0.26] relative to J ($\delta = 0.22$), with a positive RRR in each attacker budget and in both halves of the worlds, and lower regret than R on clean worlds. The two contrasts say different things. Against R , which shares the objective, horizon, cost handling, posterior, decision and spend, the gain comes from the look-ahead model, as the package of four differences listed in Sec. III. Against J , which shares the model, the gain comes from the objective: J spends budget on resolving source states that do not change the decision, and loses to C even on clean worlds (0.69 [0.62, 0.75]), so it is a weak reference rather than a strong competitor.

Where it stops. Fig. 2 and Table III show the limits, which travel with the claim. (i) The gain tracks how closely the attacker’s lies match the defender’s model: largest for cover-up, then decoy, then collude-cheap (half of whose lies are cover-ups), small for misattribution (0.05 [0.01, 0.10] relative to R). (ii) Against the adaptive trust-harvest attacker, C still beats R (0.12 [0.07, 0.18]) but not J (−0.03 [−0.09, 0.03]), and every deterministic policy loses to random acquisition (regret 0.126). The same held in E1 (random 0.125) and E2, where the score-proportional randomised J (0.115) and uniform random (0.120) were the two best. (iii) C is not the best policy on collude-cheap: nominal EIG, which ignores cost, avoids the cheap sources the attacker controls and does better. The pre-registered claim is a comparison among cost-aware policies, not “best against this attacker”.

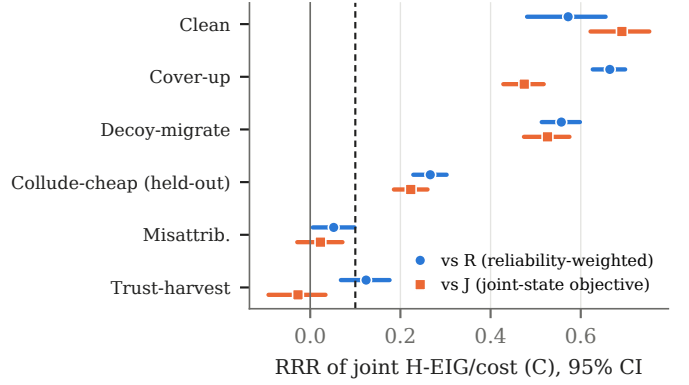


Fig. 2. E3 scope map (descriptive, not gated): RRR of C against R (circles) and J (squares) by attacker family, 95% interval. The gain is large against cover-up, decoy and colluding attackers, small against misattribution; against trust-harvest it remains relative to R but not to J .

TABLE III
E3 MEAN NORMALISED REGRET BY ATTACKER FAMILY (400 WORLDS; 537,600 POLICY-EPISODES IN TOTAL). BOLD: LOWEST IN THE COLUMN AMONG ALL SEVEN POLICIES (ONE-STEP $\hat{r}$ POLICY NOT SHOWN).

Policy	Clean	Cover	Decoy	Collude	Misattr.	Harvest
Random	0.100	0.183	0.156	0.170	0.185	0.126
EIG (nominal)	0.051	0.319	0.137	0.132	0.280	0.259
EIG- $\hat{r}$ /cost LA2 (R)	0.033	0.229	0.122	0.212	0.190	0.254
Joint (H,S)-EIG/cost (J)	0.046	0.146	0.114	0.200	0.184	0.217
Joint H-EIG/cost (C)	0.014	0.077	0.054	0.156	0.180	0.223
Decision-value probe	0.017	0.076	0.063	0.143	0.195	0.205

VI. REAL-DATA ATTEMPT: HAI

Mapping. HAI is a public hardware-in-the-loop industrial-control dataset with labelled attacks [2], [24]; several of its scenarios hold a process variable at a stale value while the process moves, a real cover-up. We used HAI 20.07 for development (detectors and model fitting, 76 episodes), HAI 21.03 for the verdict and HAI 22.04 as secondary (nine attacks whose targets the dataset manual leaves ambiguous were excluded before the run). An episode is one attack segment or a sampled normal window, as many normal windows as attacks, which sets the base rate at one half by design. Hypotheses are normal, attack confined to the boiler process, or attack touching the other processes; the three decisions keep their simulator meaning. Nine process variables and controller outputs are sources. A query reads a source’s change alarm over a 30-second window (threshold at the 99th percentile of attack-free training data). A spoofed variable stays quiet, so the alarm cannot see the spoof itself, only its absence while other sources move. A probe checks the source against a ridge regression on the other points of its process. The k -th read of a source returns the k -th window, up to six reads. The defender model was fitted on development episodes only and applied unchanged; the protocol was committed before any confirmatory episode was built, and an in-sample look at the development data was disclosed in it.

Outcome: invalid. V-real failed. On 21.03 nominal EIG

TABLE IV
E4 (HAI) MEAN NORMALISED REGRET, ALL EPISODES. V-REAL
COMPARES NOMINAL EIG WITH THE PRIOR-ONLY DECISION.

Policy	HAI 21.03	HAI 22.04
Prior only (no evidence)	0.215	0.248
Random	0.223	0.261
EIG (nominal)	0.175	0.331
EIG- $\hat{r}$ /cost LA2 (R)	0.232	0.363
Joint (H,S)-EIG/cost (J)	0.135	0.348
Joint H-EIG/cost (C)	0.210	0.358
Decision-value probe	0.105	0.297

improved on the prior-only decision by 0.040 $[-0.041, 0.119]$ (100 episodes, 50 attacks, 19 with spoofing); on 22.04 every evidence-reading policy did worse than the prior ($-0.083 [-0.172, 0.004]$). Under the pre-registered rule an invalid run decides nothing, so the comparison of belief models is untested. Table IV shows why the evidence was too weak: missed boiler attacks dominate regret, and no deterministic policy ever probed (the development probe failed 0.355 of healthy and 0.674 of spoofed readings, at twice a query’s cost). E4 therefore tested our detector more than our acquisition rules.

Exploratory observations. Had V-real passed, the E3 replication would have been killed: on 21.03 J had lower regret than C ($-0.56 [-1.44, -0.12]$), while on 22.04 the two were level ($-0.03 [-0.20, 0.11]$). On 21.03 the lowest regret came from P (0.105), which never probed and so acted as a decision-value query selector; this neither revives E1, whose kill concerned probing, nor can be read as evidence under an invalid run. We do not re-run E4; a new attempt needs a new protocol and evidence that passes V-real first.

VII. THREATS TO VALIDITY

External validity. E1–E3 use one simulator family and one world distribution with wide, uncalibrated ranges; compromise priors up to 0.40 and up to three controlled sources are far above typical tampering prevalence. Sources are conditionally independent, while real telemetry shares failure modes. The only real-data attempt was invalid, and its exploratory direction on 21.03 disagreed with E3. The positive result must be read as simulator-only.

Model match. Every attacker, held-out ones included, appears under benign activity with the probability the defender’s coupling assumes, and collude-cheap’s lies are half cover-ups. “Held out” therefore does not mean “model-mismatched”. The gain of C shrinks as the match weakens (Sec. V-B), and the same researcher wrote the policies and the attackers.

Attribution. The C -vs- R gain belongs to a package of four modelling differences, not one; the C -vs- J gain is an objective effect against a comparator that is weak even on clean worlds. Separating the four model differences, testing a joint model with a deliberately wrong compromise likelihood, and varying the assumed forge rate are needed before naming a mechanism.

Missing comparators. We did not test longer-horizon or POMDP planners, Thompson sampling or expected-error-

reduction rules, cost-aware nominal EIG in E3, or robust wrappers around C .

Exploratory origin. E3’s hypothesis came from E1 and E2 data after their verdicts. E3 is its first prospective test; a second independent test has not been done.

Construct validity. Regret is normalised per world and RRR is a ratio of sums, so worlds with large regret weigh more; per-budget, per-half results and Cliff’s δ are reported to show the effect is not driven by a few worlds. The HAI mapping (change alarms, a regression probe, three coarse hypotheses, balanced base rate) is one of many possible and was too weak.

Multiplicity. Each experiment controls its own error rate (intersection–union across comparators; Bonferroni across E2’s two candidates). Across the four experiments we report every outcome, including the failures.

VIII. CONCLUSION

Among cost-aware Bayesian acquisition rules in our simulator, the parts that mattered were the look-ahead model and the objective: modelling each source’s compromise state jointly with the hypothesis, and spending budget on information about the hypothesis rather than about the sources. That result holds against attackers whose lies resemble the defender’s model, shrinks against misattribution, and against an attacker that exploits the defender’s choices it survives only relative to reliability weighting, while random acquisition is best. It is not yet supported by real data, and the one real-data look, though invalid, pointed the other way on the objective contrast. Decision-valued probing and robust wrappers both failed pre-registered tests; of the three ideas tested in simulation, two failed.

Practical implications. Two cautious lessons follow. First, if a source can be compromised, model how it would lie rather than down-weighting it as noise, and judge evidence by what it says about the incident rather than about the sources. Second, when an attacker can watch which sources the hunter relies on, a fixed, model-driven reading order is exploitable; in every simulated experiment, randomising the order did better against such an attacker. Neither lesson replaces calibrated per-source models, which a SOC rarely has.

ARTIFACT AVAILABILITY

Code, protocols, gate code, raw outputs, statistics and verdicts are in the repository <https://github.com/ftaakash/Hades> (branch `claude/hades2-probe-pilot-gemc0j`). Experiment IDs: E1 `phase7l_probe_pilot_v2`, E2 `phase7t_robust_pilot_v1`, E3 `phase7u_joint_belief_v1`, E4 `phase7v_hai_replication_v1`. Every number in this paper is regenerated from the raw outputs by `scripts/paper/recompute_headlines.py` and `scripts/paper/make_figures.py`. HAI is available from <https://github.com/icsdataset/hai> under CC BY-SA 4.0.

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